

Effect of weaning time on the growth rate and food intake of the spiny mouse pup

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Abstract

The spiny mouse *Acomys cahirinus* produces well-developed pups although it is a relatively small mammal (45 g). We envisioned two opposing hypotheses on the effect of early weaning on the growth rate of pups. The first predicts little effect since the increase in energy intake of dams above non-reproducing females is relatively low, suggesting that pups consume a large portion of their energy as solid food, and the pups are very well developed at birth. The second predicts a substantial effect since the 'index of precociality', that is the energy intake for maintenance of a pup as a proportion of that predicted for a rodent of its body mass, falls within values for altricial rodents, suggesting an extended maternal dependence of the young. To test these hypotheses, we measured the growth rate and food intake of pups weaned after either 7, 14, 21 or 28 days. Only three of 12 pups weaned after 7 days survived and, consequently, the latter hypothesis was supported. All pups weaned at 14–28 days survived. There was a significant decrease in growth rate during the first day after weaning in pups weaned at 7, 14 and 21 days but not after 28 days, suggesting that pups did not require parental care by day 28. Peak growth rate in pups weaned at 14 days occurred in the second week but occurred in the third week in pups weaned at 21 and 28 days. In spite of these differences, pups in all treatments had similar body mass at 64 days, indicating compensatory growth. We concluded that pups of *A. cahirinus* are precocial from a morphological aspect in that they are well developed at birth but altricial from a nutritional aspect in that they require extended maternal support.

Introduction

Two strategies have evolved in the postnatal development of mammalian pups. Altricial pups are typically born naked, uncoordinated and helpless, have closed eyelids, are relatively immobile and have a long postnatal period of parental dependency. Precocial young, in contrast, are well developed, have pelage and opened eyes, are mobile shortly after birth and are associated with a lengthened gestation period.

The majority of mammal species produce altricial neonates; however, precocial young occur in many orders (Derrickson, 1992). Small mammals usually produce large litters of altricial young, resulting in high reproductive rates. In contrast, precocial neonates are generally produced by larger mammals (Derrickson, 1992) that give birth to a few young after a long gestation. For example, under good conditions, a female *Cavia aperea*, a precocial species, can produce about 10 young per year, whereas the similar-sized Norway rat *Rattus norvegicus*, an altricial species, can produce more than 100 offspring (Kraus, Trillmich & Künkele, 2005). Species producing precocial young may be less likely to reproduce seasonally than species with altricial young (Trillmich, 2000; Kraus *et al.*, 2005).

The rapid development of precocial pups includes early homeothermy, which is associated with an increase in metabolic rate (Webb & McClure, 1989; Zelditch *et al.*, 2003). As a consequence, there is generally less proportional energy intake available for growth for precocial pups than for altricial pups. The energy available for growth as a proportion of metabolizable energy intake for the two precocial species *Sigmodon hispidus* and *Cavia procellus* was 0.24 and 0.25, respectively (using the mean where one species was measured more than once). These values were substantially higher than the 0.42–0.70 for six altricial rodent species. The proportions within species were relatively constant, independent of number of pups in the litter, and there was no overlap between precocial and altricial pup species. The proportion for *Acomys cahirinus* was 0.32, that is, close to the precocial species (Kam, Khokhlova & Degen, 2006). However, when comparing the 'index of precociality' among these species, a different pattern emerged. This term, coined by Kam *et al.* (2006) represents the proportion of energy for maintenance of the pup as a proportion of that predicted for a rodent of its body mass. The calculated proportion was 0.73 and 0.93 for *S. hispidus* and *C. procellus*, respectively, but only 0.07–0.46 for the six

altricial species. The proportion for *A. cahirinus* was 0.25, which fell within the range of altricial species (Kam *et al.*, 2006).

Although development of precocial mammals is associated with high energy costs (Koteja *et al.*, 1993; Hackländer, Arnold & Ruf, 2002; Kam *et al.*, 2006), there are a number of advantages of being precocial. Precocial young achieve parental independence relatively early and are thus able to survive without their parents. In addition, young precocial small mammals possess a high level of mobility which enables them to forage more efficiently and to avoid predators. To allow these characteristics, peak growth rate occurs earlier in precocial than in altricial mammals (Barry & Kemper, 1982; Gaillard *et al.*, 1997) and maximum rate of milk production of dams occurs during early lactation in precocial rodents but during middle or late lactation in altricial rodents (Künkele & Trillmich, 1997). The energy intake in young of precocial mammal species should place less demand on females to produce milk than altricial species, due to the ability of precocial young to obtain their own solid food at an early age (Martin, 1984). Furthermore, Künkele & Trillmich (1997) hypothesized that a more evenly distributed energy demand over the course of reproduction and lactation allows precocial rodents to reproduce under conditions of low food availability, whereas altricial rodents are forced to cease reproduction.

The common spiny mouse *A. cahirinus*, a relatively small mammal (adult weighs 40–60 g), reproduces throughout the year, but mainly from the end of winter throughout spring till late summer (March–September). Gestation is long for a rodent of its size, ranging between 37 and 41 days. Peitz (1981) reported that the oestrous cycle of the spiny mouse was 11.1 days, the average age at vaginal opening was 45 days and the first litter occurred at 103 days. From these figures, females conceived for the first time at *c.* 64 days of age. They can produce litters of one to four pups two to three times per year. There is a wide range of body masses of the pups with a mean of 6.4 g on day 2 (Porter, Makin & Matochik, 1986). Pups start to consume solid food from day 2 to 3 after parturition and weaning occurs at 2–3 weeks (D'Udine, Gerosa & Drewett, 1980).

Pups are born with developed senses, their bodies are covered with short hair and they are able to move around by the second day (Derrickson, 1992; Derrickson, Jerrard & Oftedal, 1996). The prolonged gestation period results in well-developed offspring. For example, their limb muscles are more developed than those of altricial dormice *Eliomys melanurus*, voles *Microtus socialis* and jirds *Meriones tristrami* (Oron, Maltz, Shefer & Eilam, 1998) and their kidneys (Dickinson *et al.*, 2005), livers (Lamars *et al.*, 1985), lungs (Oosterhuis *et al.*, 1984) and various brain regions (Brunjes, 1989) are complete at birth. In fact, *A. cahirinus* is the first rodent species shown to complete nephrogenesis *in utero* (Dickinson *et al.*, 2005).

Lactation is very demanding energetically and is usually accompanied by a large increase in metabolizable energy intake by the lactating female. In general, the increase in energy intake by the lactating female above that of the non-

reproducing female ranges between 65 and 200% (Kaczmarewski, 1966; Migula, 1969; Randolph *et al.*, 1977; Millar, 1978; Mattingly & McClure, 1982; Glazier, 1985; Gittleman & Thompson, 1988; Poppitt, Speakman & Racey, 1994). This can be much higher in small rodents and other small mammals and can reach 400% in the house mouse *Mus musculus* (König & Markl, 1987) and 800% in a small shrew *Sorex coronatus* (Genoud & Vogel, 1990).

The increase in energy intake during lactation in the common spiny mouse, however, was somewhat lower than that found for most rodents and averaged only 45% above that of non-reproducing females (Degen *et al.*, 2002). This may indicate low milk production by the dams and would suggest that the pups are able to consume a large portion of their energy needs as solid food early and compensate for low milk intake (Degen *et al.*, 2004).

Two opposing hypotheses emerged when considering the effect of early weaning on the growth rate of *A. cahirinus* pups. In the first scenario, based on the high degree of pup development at parturition and the relatively low increase in energy intake of lactating dams, it can be hypothesized that these pups are relatively independent of maternal care and that early weaning would have little effect on their growth rate. In the second scenario, based on the relatively low index of precociality and small body size of *A. cahirinus*, it can be hypothesized that these pups are dependent on maternal care and that early weaning would have a substantial effect on their growth rate. To test these hypotheses, we measured the growth rate and food intake of spiny mice offspring after weaning at different time periods.

Materials and methods

Acomys cahirinus distribution and laboratory maintenance

The common spiny mouse *A. cahirinus* (Muridae) is a widespread nocturnal, omnivorous desert rodent (adult body mass about 45 g; Degen *et al.*, 1986) that inhabits rocky areas from eastern and northern Africa to Iran (Harrison & Bates, 1991). In Israel, it inhabits all rocky areas, including the entire Jordan Valley and the Arava (Mendelssohn & Yom-Tov, 1987) and in *wadis* near rocky outcrops (Krasnov *et al.*, 1996). These mice do not dig burrows but live in crevices among rocks (Shkolnik & Borut, 1969). *Acomys cahirinus* are readily maintained and bred in the laboratory. The progenitors of the *A. cahirinus* colonies in our laboratory were captured in the Ramon erosion cirque, Negev Highlands near Sede Boqer, Israel (30°35', 34°45'E).

Laboratory born primiparous females, aged *c.* 120 days (ranging from 105 to 135 days), were maintained individually in metabolic cages (11 × 13 × 20 cm) at a mean air temperature of 25 °C and a photoperiod of 12L:12D. They were mated by placing a male into the cage for 2 weeks, after which time the males were removed. If a female was not pregnant, a male was placed in her cage again as above. In total, 124 females raised 220 pups: 38 females raised one

pup, 76 females raised two pups and 10 females raised three pups.

Dams nursed their natural litters; there was no manipulation of litter sizes. Within litter sizes, dams were assigned randomly to one of the treatments in which pups were weaned at either 7, 14, 21 or 28-day post-parturition (four treatment groups). The latter three treatments each had 10 dams nursing one pup, 24 dams nursing two pups and three dams nursing three pups. The first treatment was curtailed after many of the pups died after weaning of 7 days (see 'Results'). We used the pups that were weaned at 14, 21 and 28 days as controls for the pups weaned at 7 days, the pups weaned at 21 and 28 days as controls for pups weaned at 14 days and pups weaned at 28 days as controls for pups weaned at 21 days. Millet seeds and alfalfa (*Medicago* sp.) were offered *ad libitum*. Drinking water was not available throughout the study as alfalfa supplied enough water. Females and offspring were weighed to 0.01 g (Model 500 PJ, Precisa, Milton Keynes, UK) daily before food was offered. Data were collected over a period of 24 months.

Weaned pups of one and two pup litters were kept individually in metabolic cages and were weighed daily till 64 days of age. This age was chosen as it is approximately the age at first conception (calculated from Peitz, 1981). Dry matter intake (DMI) was measured in all three pups that survived the 7-day weaning and in 16 individuals, eight males and eight females, per treatment that were taken randomly from a set of twins weaned at 14, 21 and 28 days and. To determine dry matter content of seeds and alfalfa, daily samples were weighed and oven dried at 100 °C to constant mass. DMI was calculated as the difference between DM offered and DM not consumed.

Statistical analyses

All data were analyzed using Statistica for Windows software (StatSoft Inc., Tulsa, OK, USA), and $P < 0.05$ was chosen as the minimal acceptable level of significance. Values are presented as means $\pm$ SD. One-way ANOVA was used to test for differences in body mass among litter sizes at birth; mean body mass of each litter was taken for litters of two and three pups as the sampling unit. Two-way ANOVA was used to test for body mass differences due to weaning time and litter size (only used one and two pups – sample size of three pups was very small and not included in the analysis). A *t*-test was used to test for differences in body mass between male and female pups at birth and at 64 days.

Results

Initial body mass of the females was 39.8 ± 5.57 g ($n = 124$). Body mass of all pups at birth was 6.25 ± 0.67 g. Pups from litters of one (6.69 ± 0.67 g; $n = 38$) were significantly heavier ($P < 0.01$) at parturition than pups from litters of three (5.62 ± 0.52 g; $n = 10$); body masses of pups from litters of two (6.26 ± 0.59 g; $n = 76$) were intermediate and did not differ significantly from either of the other litter sizes. In several cases, pups lost body mass on the first day after

parturition but increased in body mass in all cases after the second day. Body mass of male ($n = 105$) and female ($n = 115$) pups at birth did not differ (6.26 ± 0.65 and 6.23 ± 0.69 g, respectively). In addition, growth rate of male and female pups did not differ significantly, but a gender difference was found at day 64, when females ($n = 104$) weighed 30.2 ± 4.1 g and males ($n = 101$) weighed 33.5 ± 4.7 g ($P < 0.01$).

Of the 12 pups separated from their mothers after 7 days, only three pups survived. Two pups died between 8 and 14 days of age (first week after weaning), four pups between 15 and 21 days of age and one pup between 22 and 28 days of age. DMI for the first 3 days of the nine pups that died was $2.55 (\pm 0.45) \text{ g kg}^{-0.75} \text{ day}^{-1}$ while that of the three that survived was $15.49 (\pm 5.46) \text{ g kg}^{-0.75} \text{ day}^{-1}$. Among survivors, two pups were born in litters of one and one pup from a litter of two. Pups that survived were significantly ($P < 0.05$) heavier at parturition ($n = 3$; 7.33 ± 0.51 g) than pups which died after weaning ($n = 9$; 6.46 ± 0.44 g) and started increasing their body masses from day 12 or 13 post-parturition (Fig. 1). We realized that most of the pups could not survive weaning at 7 days and, therefore, stopped this treatment after 12 pups were weaned.

Pups weaned at 14 and 21 days survived, but there was a significant decrease ($P < 0.01$) in growth rate (Table 1) on the first day after weaning compared with non-weaned pups. Growth rate of pups weaned at 14 days (Fig. 2) peaked in the second week and earlier than in pups weaned at 21 days (Fig. 3) and 28 days (Fig. 4), which peaked in the third week. Pups weaned at 28 days had the same body mass increase on day 29 as pups weaned at 14 and 21 days. Mean body mass at 64 days did not differ between litters of one and two pups and among treatments and averaged 35.1 ± 8.7 g ($n = 3$) for 7-day weaning, 32.5 ± 4.9 g ($n = 33$) for 14-day weaning, 31.9 ± 4.3 g ($n = 33$) for 21-day weaning and 31.5 ± 4.7 g ($n = 33$) for 28-day weaning.

In the first 3-day post-weaning, DMI of pups weaned at 7 days was $15.5 \text{ g kg}^{-0.75} \text{ day}^{-1}$, which was substantially lower ($P < 0.05$) than the $39.6\text{--}43.4 \text{ g kg}^{-0.75} \text{ day}^{-1}$ for pups weaned 14–28 days (Table 2). Differences between the four

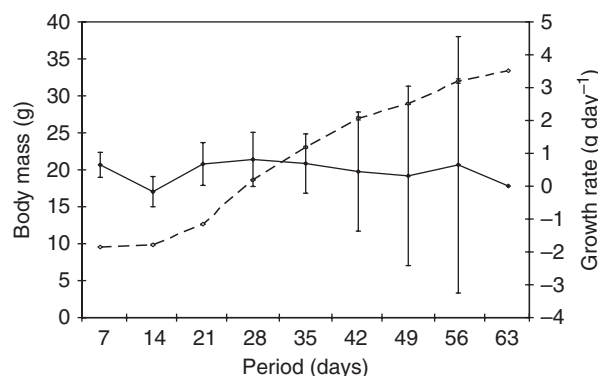


Figure 1 Body mass (broken line) and daily growth rate (solid line) over time of *Acomys cahirinus* pups that survived when weaned at 7 days ($n = 3$).

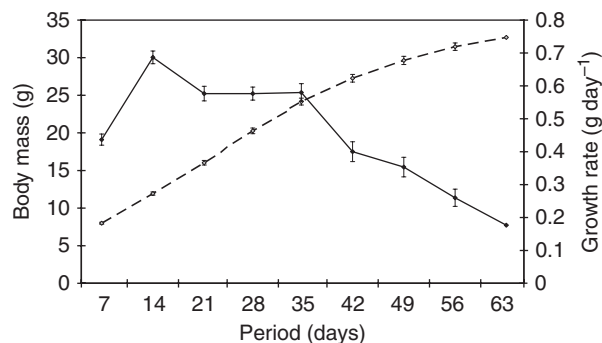
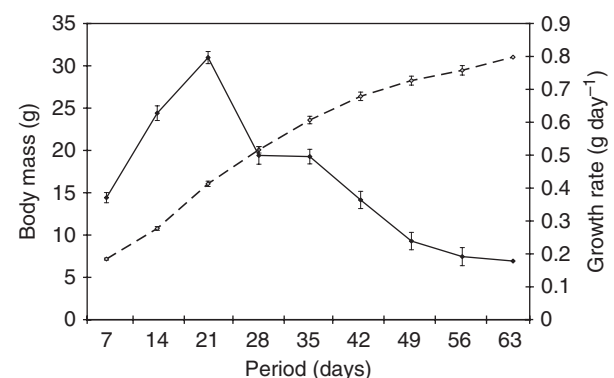
Table 1 Effect of weaning time (days after birth) and age (in days) on the growth rate ($\text{g day}^{-1} \pm \text{sd}$) of *Acomys cahirinus* pups 1 day after weaning

<i>n</i>	Weaning	Age (days)	Growth rate	<i>P</i> ^a
12	7	8	-0.71 ± 0.30	*
105	14, 21, 28	8	0.49 ± 0.28	
10	7	15	-0.17 ± 0.43	*
35	14	15	0.19 ± 0.57	*
70	21, 28	15	0.80 ± 0.36	
4	7	22	0.67 ± 0.68	NS
35	14	22	0.47 ± 0.51	NS
35	21	22	0.35 ± 0.51	*
35	28	22	0.87 ± 0.42	
3	7	29	0.53 ± 0.28	NS
35	14	29	0.50 ± 0.46	NS
35	21	29	0.58 ± 0.60	NS
35	28	29	0.58 ± 0.60	

^aStatistical comparison between weaned and non-weaned pups. Last section is between pups weaned at 28 days versus the others weaned earlier.

* $P < 0.05$.

n, number of litters with mean body mass per litter, except for 7-day weaning where all pups are presented; NS, not significant.

**Figure 2** Body mass (broken line) and daily growth rate (solid line) over time of *Acomys cahirinus* pups weaned at 14 days ($n=35$).**Figure 3** Body mass (broken line) and daily growth rate (solid line) over time of *Acomys cahirinus* pups weaned at 21 days ($n=35$).

groups, however, were marginal in days 4–7 post-weaning and ranged between 39.4 and $44.0 \text{ g kg}^{-0.75} \text{ day}^{-1}$. The proportional increase in intake between days 4 and 7 post-weaning compared with 1–3-days post-weaning decreased ($P < 0.05$) with an increase in weaning time. It was nearly three times the intake in pups weaned after 7 days and near to zero increase in pups weaned after 28 days (Table 2).

Discussion

We presented two plausible opposing hypotheses on the effect of early weaning on the growth rate of *A. cahirinus* pups; in the first there would be little effect and in the second there would be substantial effect. In general, the pups were not able to survive the second week of life without maternal care and, consequently, our study supported the latter hypothesis. We concluded that pups of *A. cahirinus* are precocial from a morphological aspect in that they are well developed at birth but altricial from a nutritional aspect in that they require extended maternal support.

Only three of the 12 pups weaned at 7 days survived. The three survivors were heavier than non-survivors at weaning and all three pups lost body mass after weaning. The daily DMI in the first 3 days after weaning of the pups that died was only 16% of the intake of the surviving pups, and it was evident that their energy intake without milk was insufficient to sustain them. Even the intake of the three pups that survived was extremely low in the first 3 days after weaning, only about 33% of required intake for maintenance and growth (taking about $45 \text{ g kg}^{-0.75} \text{ day}^{-1}$ as required intake; Table 2), but perhaps their larger body mass allowed them to survive this vulnerable period. Our results were in agreement with the study of D'Udine *et al.* (1980) who reported that nursing was substantial within the first week of age and occurred above 40% of the time at 5–8 days after birth, but decreased in the second and third weeks to about 25% of the time.

Pups weaned after 14 and 21 days lost body mass the first day after weaning which would indicate that the pups were still dependent on their mothers' milk for a portion of their

energy intake till age 21 days. However, pups weaned at 28 days increased their body mass the day after weaning at the same rate as pups which were weaned after 21 days, which indicated that they were consuming solid food equivalent to their energy needs and were independent of milk. In addition, pups weaned at 28 days had a peak growth rate during the third week of lactation, that is previous to weaning, which indicated that this was the normal occurrence. Pups weaned at 21 days also had their peak in the third week, however pups weaned at 14 days had their peak growth rate in their second week, which indicated that the weaning was before reaching maximum growth rate.

Age of first solid food intake in mammals generally ranges between 14 and 50 days (Hayssen, 1993). However, solid food is eaten within 1 week after birth, and occasionally on the day of birth, in hystricomorph rodents such as guinea pigs, porcupines, coypus and agoutis, as well as in other mammals such as sloths, zebras, hyraxes and bovines. In contrast, first solid food is eaten near weaning in many squirrels and murid rodents.

Acomys cahirinus pups were not able to consume adequate solid food immediately after birth. Some precocial mammals are able to do so. For example, guinea pig *C. procellus* offspring survived when weaned at 5 days of age, although lactation in the domestic guinea pigs lasts about 23 days. They began to feed on solid food immediately after birth and consumed a considerable amount of

food within 3-day post-partum. During the first 20 days of lactation, nearly 60% of the metabolizable energy intake of the pups was from solid food. Pups deprived of solid food grew slower and were in poorer body condition than pups in control litters, due to an inadequate energy intake. *Cavia procellus* dams did not increase energy consumption or milk production and did not increase energy allocation to pups, even though their offspring were in poor condition (Künkele & Trillmich, 1997; Laurien-Kehnen & Trillmich, 2003).

In spite of the different responses to weaning time, body mass at 64 days did not differ among treatments, indicating the potential compensatory growth of the pups. The pups weaned at 7 days had the lowest food intake and largest body mass loss immediately after weaning. However, the increase in their intake days 4–7 post-weaning compared with days 1–3 was almost threefold and the highest proportional increase of any of the weaned groups (Table 2). Furthermore, this group showed the highest intakes of any of the groups between 14 and 28 days of age, considerably higher than the other groups. We are aware that there were only three individuals in this group, but the trend of compensatory growth was apparent.

Compensatory growth occurs in many animal species, both poikilotherms and homeotherms. This phenomenon is characterized by a period of slow growth or even a decrease in body mass followed by a period of accelerated growth. The delay in growth is commonly associated with a nutritional deficit and the compensation occurs when conditions improve. Although the growth process resulted in a similar final body mass among treatments in our study, the delayed growth may have a negative long-term effect, especially, under natural conditions. For example, it may lead to increased predation risk, lower survival, lower dominance rank and, hence, decreased fitness (Metcalf & Monaghan, 2001).

In general, 'bad starts' related to nutritional shortages are due to either insufficient time to acquire resources or are associated with poor parental provisioning (Metcalf & Monaghan, 2001). Weaned pups in our study did not face a food shortage or feed of poor quality but rather faced independent life without milk and maternal care. It is unknown whether one of the two factors or a combination of both played a crucial role during the first 2 weeks of age. Surprisingly, a delayed growth was evident even for pups weaned after 21 days. The increased feeding rate 4–7-day

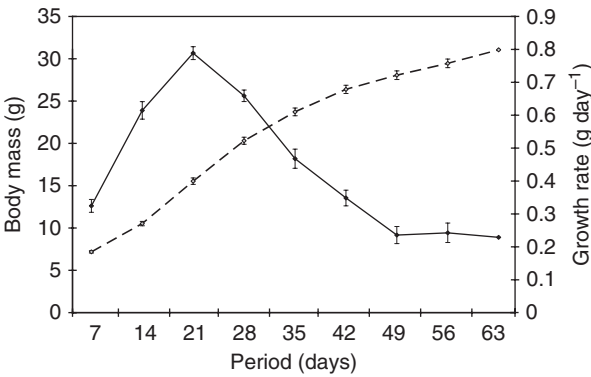


Figure 4 Body mass (broken line) and daily growth rate (solid line) over time of *Acomys cahirinus* pups weaned at 28 days ($n=35$).

Table 2 Mean ($\pm$ SE) DMI ($\text{g kg}^{-0.75} \text{ day}^{-1}$) in post-weaned *Acomys cahirinus* offspring at different ages

Weaning day	<i>n</i>	Days post-weaning		Increased intake	Age (days)			
		1–3	4–7		7–13	14–20	21–27	28–34
7	3	15.5 $\pm$ 5.5	39.4 $\pm$ 3.9	2.78 $\pm$ 0.13 ^a	29.3 $\pm$ 2.2	49.5 $\pm$ 5.5	57.0 $\pm$ 3.2	47.9 $\pm$ 0.9
14	16	39.6 $\pm$ 4.4	48.1 $\pm$ 3.3	1.30 $\pm$ 0.07 ^b		43.7 $\pm$ 1.6	47.9 $\pm$ 1.3	43.8 $\pm$ 2.4
21	16	40.4 $\pm$ 2.7	44.7 $\pm$ 2.1	1.13 $\pm$ 0.04 ^c			42.7 $\pm$ 1.4	46.2 $\pm$ 1.3
28	16	43.4 $\pm$ 3.0	44.0 $\pm$ 3.1	1.03 $\pm$ 0.04 ^c				44.0 $\pm$ 2.1

The increased intake ratio during first week post-weaning, calculated as the DMI ratio of 4–7 days to 1–3-day post-weaning, included. Means with different superscripts are different from each other.

DMI, dry matter intake.

post-weaning compared with that of 1–3 days may indicate the volume of milk consumption before weaning. It may also reflect higher energy expenditure of the pup; huddling with its mother and its siblings, which may lower energy needs for maintenance, was not available.

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